

Cohomological rigidity of simple polytopes

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This presentation is based on the work jointly with

Taras Panov (Moscow State Univ.)

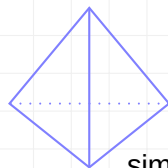
Dong Youp Suh (KAIST)

You can see the paper on the web [arXiv:0807.4800](https://arxiv.org/abs/0807.4800).

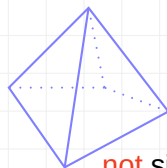
Simple polytope

A **(convex) polytope** is the convex hull of a finite set of points in some $\mathbb{R}^n$. Two polytopes are **combinatorially equivalent** if there is a bijection between their sets of faces that preserves the inclusion relation.

An n -dim. polytope is **simple** if each vertex is the intersection of exactly n facets, i.e. faces of dim. $n - 1$.



simple



not simple

Polytope

Let $P \subset \mathbb{R}^n$ be a convex simple n -polytope with m facets. Then P is an intersection of finitely many half-spaces in some $\mathbb{R}^n$:

$$P = \{\mathbf{x} \in \mathbb{R}^n : \langle \ell_i, \mathbf{x} \rangle \geq a_i, i = 1, \dots, m\},$$

where $\ell_i \in (\mathbb{R}^n)^*$ are some linear functions and $a_i \in \mathbb{R}$ for all i . Form the $n \times m$ -matrix L whose columns are the vectors ℓ_i , i.e., $L_{ji} = (\ell_i)_j$ and let $\mathbf{a} = -(a_1, \dots, a_m)^t$. Then

$$P = \{\mathbf{x} \in \mathbb{R}^n : (L^t \mathbf{x} + \mathbf{a})_i \geq 0, i = 1, \dots, m\}.$$

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Consider the affine map

$$A_P : \mathbb{R}^n \rightarrow \mathbb{R}^m, A_P(\mathbf{x}) = L^t \mathbf{x} + \mathbf{a} \in \mathbb{R}^m.$$

Then $P \subset \mathbb{R}^n$ and $A_P(P) \subset \mathbb{R}_{\geq 0}^m$ are affinely equivalent.

Moment-angle manifold of simple polytope

The **moment-angle manifold** $\mathcal{Z}_P$ is defined by the pull-back diagram

$$\begin{array}{ccc} \mathcal{Z}_P & \xrightarrow{i_Z} & \mathbb{C}^m \\ \varrho_P \downarrow & & \downarrow \varrho \\ P & \xrightarrow{A_P} & \mathbb{R}_{\geq 0}^m \end{array}$$

Here $\varrho(z_1, \dots, z_m)$ is given by $(|z_1|^2, \dots, |z_m|^2)$.

$\mathcal{Z}_P$ is *only* depend on the **combinatorial structure** of P .

Face ring of simple polytopes

- P : n -simple polytope with m facets
- $\mathfrak{F}(P) = \{F_1, \dots, F_m\}$: set of facets

The **face ring** (or the **Stanley-Reisner ring**) of P is the quotient ring

$$\mathbb{Q}(P) = \mathbb{Q}[v_1, \dots, v_m] / \mathcal{I}_P,$$

where $\mathcal{I}_P$ is the homogeneous ideal generated by all square-free monomials $v_{i_1} \cdots v_{i_s}$ such that $\bigcap_{\ell=1}^s F_{i_\ell} = \emptyset$.

W.Bruns, J.Gubeladze (1996)

Let P and Q be two n -simple polytopes.

$$\mathbb{Q}(P) \cong \mathbb{Q}(Q) \text{ as } \mathbb{Q}\text{-algebras} \iff P \approx Q$$

Tor-algebra

- A : finitely generated graded algebra
- M, N : finitely generated A -modules

A **free resolution** $[R, d]$ of M is an exact sequence

$$0 \longrightarrow R^{-n} \xrightarrow{d} R^{-n+1} \xrightarrow{d} \cdots \xrightarrow{d} R^0 \xrightarrow{d} M \longrightarrow 0 ,$$

Take $\otimes_A N$ to get

$$0 \longrightarrow R^{-n} \otimes_A N \xrightarrow{d \otimes 1} R^{-n+1} \otimes_A N \xrightarrow{d \otimes 1} \cdots \xrightarrow{d \otimes 1} R^0 \otimes_A N \longrightarrow 0$$

Then $\mathrm{Tor}_A^{-i}(M, N)$ is defined to be

$$\mathrm{Tor}_A^{-i}(M, N) := H^{-i}(R^{-*} \otimes_A N).$$

Since everything is graded,

$$\mathrm{Tor}_A^{-i}(M, N) = \bigoplus_j \mathrm{Tor}_A^{-i,j}(M, N)$$

Tor-algebra of simple polytope

- $A := \mathbb{Q}[v_1, \dots, v_m]$: the polynomial graded ring

Note that both $\mathbb{Q}(P)$ and $\mathbb{Q}$ are graded A -modules. Then we can define $\text{Tor}_A(\mathbb{Q}(P), \mathbb{Q})$.

V.Buchstaber, T.Panov(2003)

$$H^{**}(\mathcal{Z}_P, \mathbb{Q}) \cong \text{Tor}_A(\mathbb{Q}(P), \mathbb{Q}) \text{ as } \mathbb{Q} \text{ algebras,}$$

where $A = \mathbb{Q}[v_1, \dots, v_m]$.

Cohomological rigidity of simple polytope

When does $H^{**}(\mathcal{Z}_P, \mathbb{Q}) \cong H^{**}(\mathcal{Z}_Q, \mathbb{Q})$ imply $P \approx Q$?

P is (cohomologically) rigid if whenever there is another polytope Q with $H^{**}(\mathcal{Z}_P, \mathbb{Q}) \cong H^{**}(\mathcal{Z}_Q, \mathbb{Q})$, there is a combinatorial equivalence $P \approx Q$.

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Which polytopes are rigid?

Why study rigidity? : Related on Topology

The **toric manifold** is a compact non-singular toric variety of dim $2n$ on which an algebraic torus $(\mathbb{C}^*)^n$ -action with a dense orbit.

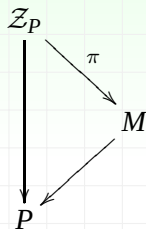
- $T^n = \{z \in (\mathbb{C}^*)^n : |z| = 1\}$: standard subgroup of $(\mathbb{C}^*)^n$.

The orbit space of a toric manifold with T^n can be identified with the simple polytope P and the action is *locally standard*. we can define the notion of a topological generalization of a toric manifold.

A **quasitoric manifold** is a smooth closed manifold with a locally standard T^n -action which has a simple polytope as an orbit space.

Toric manifolds

We get the quasitoric manifold M over P from the moment-angle manifold via π .



S.Choi, T.Panov, D.Suh (2008)

- M, N : two quasitoric manifolds over P and Q , resp.

$$H^*(M) \cong H^*(N) \implies \mathrm{Tor}(\mathbb{Q}(P), \mathbb{Q}) \cong \mathrm{Tor}(\mathbb{Q}(Q), \mathbb{Q})$$

Toric topology and Combinatorics

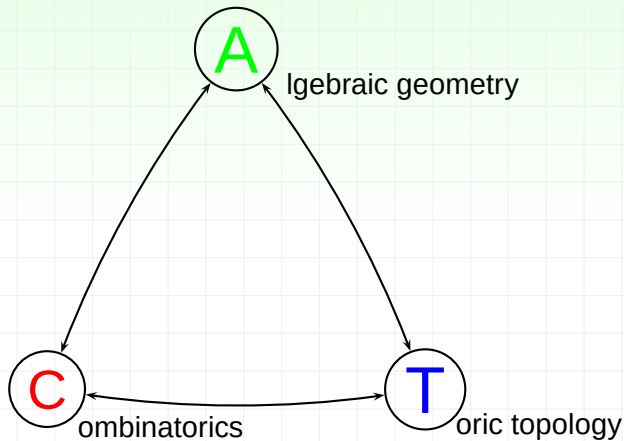
In general,

- 1 the information of Lie-group action on M is depend on the symmetry of M , that is, orbit space.
- 2 cohomology ring of M is defined independently the Lie-group action on M .

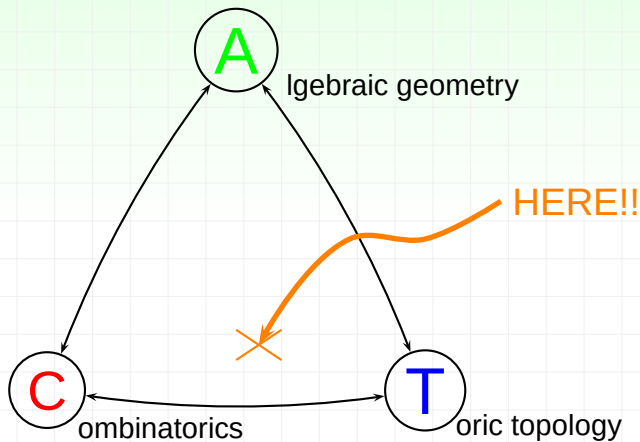
But!

If P is rigid, then the combinatorial structure of P is determined by the cohomology ring of supporting quasitoric manifold M .

This work is located



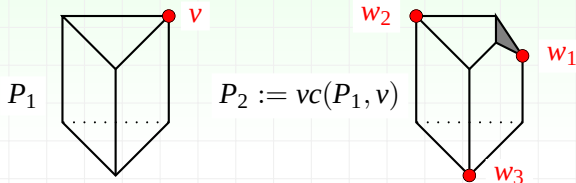
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Example of non-rigid simple polytope

Not all polytopes are rigid:

- $vc(P, v)$ = the vertex cut of P at v .



$$M_1 = \mathbb{CP}^2 \times \mathbb{CP}^1$$

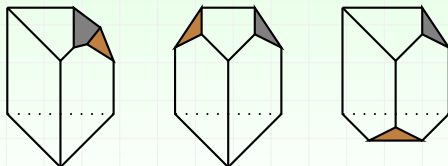
$$\begin{array}{c} \pi \downarrow \\ P_1 \end{array}$$

$$M_2 = \text{blow up of } M_1 \text{ at } \pi^{-1}(v)$$

$$\begin{array}{c} \pi' \downarrow \\ P_2 \end{array}$$

Example of non-rigid simple polytope

$$P_{3,i} := vc(P_2, w_i)$$



$$M_{3,i} = \text{blow up of } M_2 \text{ at } \pi'^{-1}(w_i)$$



$$P_{3,i}$$

$$M_{3,i} \cong (\mathbb{CP}^2 \times \mathbb{CP}^1) \# \mathbb{CP}^3 \# \mathbb{CP}^3$$

Combinatorics of Bigraded Betti number

The bigraded Betti number $\beta^{-i,j}(P)$ of P is defined by

$$\beta^{-i,j}(P) = \dim_{\mathbb{Q}} \operatorname{Tor}_A^{-i,j}(\mathbb{Q}(P), \mathbb{Q}).$$

Bigraded Betti numbers are combinatorial invariants of P !

To show uniqueness of the bigraded Betti number of P is one of the ways to decide the rigidity of P .

Combinatorics of Bigraded Betti number

M.Hochster (1977)

Let P be a simple convex polytope with facets $F_1, \dots, F_m$. For a subset $\sigma \subset \{1, \dots, m\}$ let $P_\sigma = \cup_{i \in \sigma} F_i \subset P$. Then we have

$$\beta^{-i,j}(P) = \sum_{|\sigma|=j} \dim \tilde{H}^{j-i-1}(P_\sigma).$$

Here $\dim \tilde{H}^{j-i-1}(\emptyset) = 1$ by convention.

[Example]

- $\beta^{-1,2}(P) = \#$ of pair of non-intersecting facets of P .
- $\beta^{-1,3}(P) = \#$ of 3-belts of facets of P .

Rigid polytopes

The following polytopes are rigid ;

M.Masuda, T.Panov (2007)

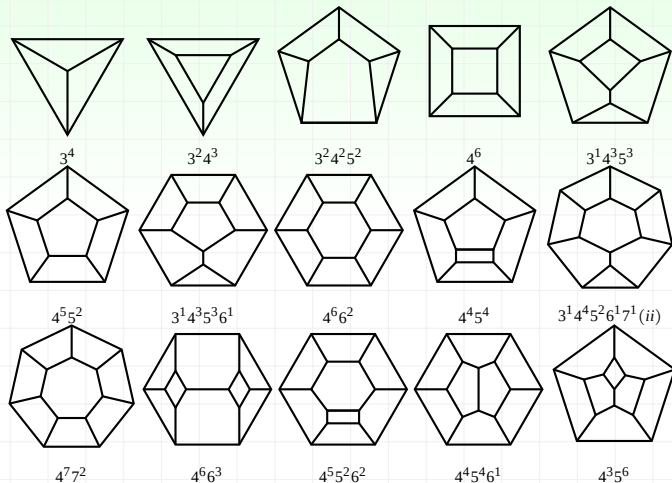
- every cube I^n .

S.Choi, T.Panov, D.Suh (2008)

- every 2-dim polytope, i.e., polygon,
- every triangle-free n -dim simple polytope with facet numbers $\leq 2n + 2$,
- any product of simplices $\prod_{i=1}^k \Delta^{n_i}$,
- vertex cut of a product of simplices $vc(\prod_{i=1}^k \Delta^{n_i})$,
- Dodecahedron.

Rigid polytopes

Rigid 3-polytopes up to 9 facets.



Further results

S.Choi

- every simple 4-polytope with 7 facets,
- vertex cut of a dodecahedron

S.Choi, J.S.Kim

- every 3-prism, i.e., $\text{polygon} \times I$.
- vertex cut of a 3-prism.

Sketch of Proof - a product of simplices

- $\sigma(P) = \sum_{j \geq 2} j \beta^{-1,j}$

$\sigma(P)$ is nothing but the sum of the degrees of all elements of a minimal basis of I_P .

[Lemma]

$$\sigma(P) = m \iff P = \prod \Delta^{n_i}$$

(Proof of lemma)

$$\begin{aligned}\mathbb{Q}(P) &\cong \mathbb{Q}[x_1, \dots, x_m]/I_P \\ &\cong \bigotimes_{i=1}^t \mathbb{Q}[y_{i,0}, \dots, y_{i,n_i}]/(g_i) \\ &\cong \bigotimes_{i=1}^t \mathbb{Q}(\Delta^{n_i}) \cong \mathbb{Q}\left(\prod_{i=1}^t \Delta^{n_i}\right).\end{aligned}$$



Sketch of Proof - a 3-prism

[Theorem] (S.Choi, J.S.Kim)

- P : 3-dim simple polytope with m -facets
- Q_k : k -gonal prism, i.e., k -gon $\times I$.
- S : stack polytope with m -facets

$$\beta^{-2,4}(P) \leq \beta^{-2,4}(S) = (m-6) \binom{m-3}{2}$$

Moreover, if $\beta^{-1,3}(P) = 0$, then

$$\beta^{-2,4}(P) \leq \beta^{-2,4}(Q_{m-2}) = \frac{(m-2)(m-5)}{2},$$

and **the equality** holds for P is stack polytope and $P = Q_{m-2}$, respectively.

Thank you for listening!